

Trust Before You Read: A Four-Case Framework for Detecting Unreliable Post-Hoc Explanations Under Real-World Data Perturbations

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Abstract—Post-hoc explanation methods such as LIME [1] and SHAP [2] are widely used to interpret machine-learning predictions. However, in real-world deployments, data frequently contains noise, missing values, and distribution shifts. Under these conditions, explanations may change significantly even when predictions remain identical, creating a silent trust failure that misleads practitioners. We present a reliability layer over existing explainers that detects untrustworthy explanations through a four-case decision framework: *RELIABLE*, *MISLEADING*, *DANGEROUS*, and *NORMAL_DRIFT*. We evaluate our framework across four real-world datasets (UCI HAR, PhysioNet EEG Microstates, Norwegian Electricity, and CIFAR-10) under four perturbation types (Gaussian noise, missing values, distribution shift, and feature corruption) using LIME and SHAP as explainers. Quantitative reliability is measured via the Explanation Reliability Index (ERI) [4] sub-scores ERI-S, ERI-R, ERI-T, and ERI-M. Our framework achieves macro F1 = 1.00 on synthetic ground-truth injection tests. ERI-S scores range from 0.591 to 0.768 across datasets, confirming that temporal structure (Norwegian LSTM) produces less stable explanations under distribution shift than high-dimensional activity recognition (UCI HAR). Model accuracies are: UCI HAR LR 95.55%, UCI HAR MLP 95.45%, EEG LR 62.07%, CIFAR-10 ResNet-18 70.01%. All code and datasets are publicly available.

Index Terms—Explainable AI, LIME, SHAP, explanation reliability, perturbation, trust framework, post-hoc explanation, ERI metric

I. INTRODUCTION

The growing deployment of machine-learning models in high-stakes domains such as healthcare, finance, and autonomous systems has created an urgent demand for trustworthy explanations. Post-hoc explanation methods, notably LIME [1] and SHAP [2], have become the standard tools for interpreting individual predictions by identifying which input features most influenced a model’s output.

Despite their widespread adoption, a fundamental question remains unanswered in practice: *can a given explanation be trusted?* An explanation is useful only if it is stable—that is, if small, realistic changes to the input produce consistent feature attributions. In real-world settings, input data is rarely clean. Sensor readings contain measurement noise, data pipelines introduce missing values, and production data drifts from

training distributions over time. Under these conditions, LIME and SHAP explanations may change substantially even when the model prediction does not change. This creates a silent trust failure: a user who sees a stable prediction assumes the explanation is stable too, but it is not.

We identify four qualitatively distinct failure modes, which we formalise as a *four-case decision framework*:

- 1) **RELIABLE**: prediction stable, explanation stable — accept the explanation.
- 2) **MISLEADING**: prediction stable, explanation changed — reject the explanation; it is sensitive to minor input perturbations.
- 3) **DANGEROUS**: prediction changed, explanation looks the same — warn the user; the explanation has hidden a real decision change.
- 4) **NORMAL_DRIFT**: both prediction and explanation changed — accept cautiously; this is expected behaviour under large perturbations.

This framework constitutes a quality-control layer that can be placed over any existing explainer without modifying the underlying model or explanation method. Our key contributions are:

- A formal four-case trust decision framework linking explanation stability to prediction correctness under realistic data perturbations.
- An empirical evaluation across four real-world datasets and four perturbation types using both LIME and SHAP.
- Adoption and extension of the Explanation Reliability Index (ERI) [4] sub-scores (ERI-S, ERI-R, ERI-T, ERI-M) as quantitative reliability metrics.
- A synthetic injection validation demonstrating that the framework classifies all four cases with macro F1 = 1.00.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the framework methodology. Section IV presents the reliability framework in detail. Section VI presents experimental results. Section VII discusses findings and limitations. Section VIII concludes.

II. RELATED WORK

A. Post-Hoc Explanation Methods

LIME [1] fits a sparse linear surrogate to a stochastic neighbourhood around each query point. Because sampling is random, explanations inherit variance proportional to neighbourhood density—extremely low in high-dimensional spaces such as the 561-feature UCI HAR dataset. SHAP [2] computes Shapley values satisfying four game-theoretic axioms (efficiency, symmetry, dummy, additivity). TreeSHAP [3] achieves exact computation for tree ensembles. GradientSHAP and DeepSHAP [2] extend attribution to neural networks via expected gradients. The fundamental local-versus-global divide between LIME and SHAP explains why they produce fully disjoint top-5 feature sets on UCI HAR and EEG Microstates, yet agree strongly (4/5 overlap) on Norwegian Electricity where a dominant recency effect creates a shared anchor feature [3].

B. Stability and Robustness of Explanations

Alvarez-Melis and Jaakkola [5] define explanation robustness as the Lipschitz constant of the input-to-explanation map, showing that standard LIME kernel settings fail this criterion. Yeh et al. [6] empirically demonstrate that high-fidelity explanations are not necessarily stable ones. Slack et al. [7] show that adversarial perturbations can fool LIME and SHAP into arbitrary explanations while preserving model predictions—the same structural decoupling of prediction and explanation stability that motivates our (δ_p, δ_e) flags. Ghassemi et al. [8] document how unstable explanations in clinical settings produce false confidence in practitioners, precisely the scenario our MISLEADING case captures. Samek et al. [9] evaluate explanation quality via pixel-flipping; their oracle-removal protocol is inapplicable in production without retraining, a limitation our framework avoids by requiring only two additional inference calls per sample.

C. The ERI Base Paper and Our Extension

Sengupta et al. [4] introduce the Explanation Reliability Index, a family of metrics measuring explanation drift under four axioms: ERI-S (sensitivity to small input perturbations), ERI-R (robustness under feature redundancy), ERI-T (temporal smoothness for sequential models), and ERI-M (consistency across model training checkpoints). Each is defined as $\text{ERI} = 1/(1 + \Delta)$ where Δ is the mean ℓ_2 drift of the explanation vector over 30 repeated perturbation trials. Evaluated on SHAP and Integrated Gradients, Sengupta et al. reveal widespread reliability failures—explanations frequently change under conditions that leave model accuracy intact.

Our work directly extends ERI. Its key limitation is that ERI produces a continuous scalar with no decision boundary: a practitioner reading $\text{ERI-S} = 0.591$ receives no guidance on whether to trust or reject the explanation. We bridge this gap by conditioning on δ_p . When ERI-S is low and $\delta_p = 0$, the case is MISLEADING and the explanation must be rejected. When ERI-S is low and $\delta_p = 1$, the case is NORMAL DRIFT and the explanation can be accepted cautiously. This

transforms a researcher-facing metric into an operator-facing action. We also extend ERI’s perturbation repertoire by adding feature corruption—not covered in the original paper—to the evaluation protocol.

D. Perturbation Protocols and Applications

The ROAR benchmark [10] evaluates explanation quality by removing top-ranked features and measuring accuracy drop, requiring full retraining. Wei et al. [11] revisit post-hoc robustness under adversarial and natural perturbations and confirm that stability varies strongly by model family and feature dimensionality. Agarwal et al. [12] survey reliable post-hoc explanations and conclude that stability under realistic data-quality perturbations remains an open gap. In high-stakes applications, Lundberg et al. [3] apply TreeSHAP to ICU mortality predictions and note that SHAP values shift significantly under background dataset changes—analogueous to our distribution shift perturbation. Regulatory frameworks including the EU AI Act and the US NIST AI RMF require documentation of explanation consistency; our framework’s (δ_p, δ_e) audit log provides structured evidence that satisfies these requirements without modifying the underlying model [12].

E. LIME Explanation Behaviour Under Perturbation

LIME is evaluated as a `LimeTabularExplainer` on UCI HAR and EEG Microstates (classification mode) and as a regression-mode explainer on Norwegian Electricity. For CIFAR-10, LIME operates on superpixel segmentations produced by the SLIC algorithm with $n_segments = 20$. In all cases, we draw 500 neighbourhood samples per explanation call (`num_samples=500`) and return the top- $k = 5$ features by absolute weight.

1) *UCI HAR — Gaussian Noise*: Fig. 1 illustrates the LIME instability on UCI HAR. The top-5 feature list before perturbation (`feat_287`, `feat_345`, `feat_547`, `feat_54`, `feat_441`) shares *zero* features with the post-perturbation list (`feat_483`, `feat_201`, `feat_197`, `feat_190`, `feat_325`). Over 10 random seeds and 20 test samples, the mean LIME top-5 overlap for UCI HAR under $\sigma = 0.2$ Gaussian noise is 0.013 ± 0.006 (95% CI: $[0.007, 0.020]$). UCI HAR has 561 features, most of which are correlated time-domain and frequency-domain derivatives of a small number of raw sensor channels. Adding $\sigma = 0.2$ noise moves the query point into a different neighbourhood region where an entirely different sparse linear model fits the local geometry, yet the logistic regression decision boundary is not crossed. The result is the MISLEADING classification: stable prediction, completely unstable explanation.

2) *EEG Microstates — Missing Values*: LIME behaves more stably on EEG Microstates. The 64 PCA-compressed features are significantly lower-dimensional than the HAR feature space. Under 15% random missingness, LIME achieves a top-5 overlap of 3/5, exceeding the stability threshold $\tau = 0.5$, and the prediction is also stable, so the trust decision is **RELIABLE**.

3) *Norwegian Electricity — Distribution Shift*: For the LSTM electricity model, LIME operates on the flattened sequence representation (120 features: 24 time steps $\times$ 5 zones) in regression mode. After a half-standard-deviation distribution shift ($s = 0.5$), the top-5 overlap drops to 1/5. Although the prediction is unchanged, the explanation has changed substantially, classifying as **MISLEADING**. This highlights a specific vulnerability of LIME under global offset perturbations: a uniform shift can still alter the local geometry by changing which background samples are nearest-neighbours.

F. SHAP Explanation Behaviour Under Perturbation

SHAP is evaluated using LinearExplainer for the logistic regression models (UCI HAR and EEG), KernelExplainer with 100 background samples and 200 perturbation samples ($n_{\text{samples}}=200$) for the LSTM, and GradientExplainer with 50 background images for ResNet-18.

1) *Agreement with LIME by Dataset*: Table I summarises per-dataset agreement between LIME and SHAP on clean data.

TABLE I
LIME-SHAP TOP-5 AGREEMENT ON CLEAN DATA

Dataset	Mean Overlap	N	Interpretation
UCI HAR	0.00 / 5	5	Complete disagreement
EEG Microstates	0.00 / 5	5	Complete disagreement
Norwegian Elec.	4.00 / 5	3	Strong agreement
CIFAR-10	0.67 / 5	3	Partial spatial overlap

The 0.00 overlap on UCI HAR and EEG Microstates reflects the fundamental design difference between the two methods [1], [2]. For Norwegian Electricity, the strong agreement (4.0/5) arises because the LSTM has a pronounced recency bias: the last 3–5 timesteps dominate the prediction and both methods converge on these regardless of whether attribution is local or global.

2) *SHAP on CIFAR-10 (GradientExplainer)*: For ResNet-18 on CIFAR-10 [16], GradientExplainer computes expected gradients with respect to a 50-image background batch. The spatial overlap between LIME superpixels and SHAP 4×4 blocks is 0.67 for images 0 and 1, and 0.50 for image 2. LIME tends to highlight contiguous object-shaped superpixels, while SHAP highlights finer-grained pixel regions tracking high-gradient transitions. The CIFAR-10 case is classified as **NORMAL_DRIFT** under pixel noise ($\sigma = 0.3$): the model reclassifies a sample from *ship* to *airplane* and the attribution map shifts accordingly—the honest expected behaviour under large-magnitude perturbation.

3) *Impact of SHAP Background Dataset on Stability*: SHAP’s KernelExplainer stability on Norwegian Electricity depends significantly on background dataset size. With 100 background samples, the top-5 SHAP features are stable across three evaluated sequences, with `zone0_t23` as the top feature in all three. Reducing to 10 background samples lowers this stability to 2/5 overlap, implying that operators should allocate at least 50–100 background samples and treat attributions from smaller sets as potentially **MISLEADING** [3].

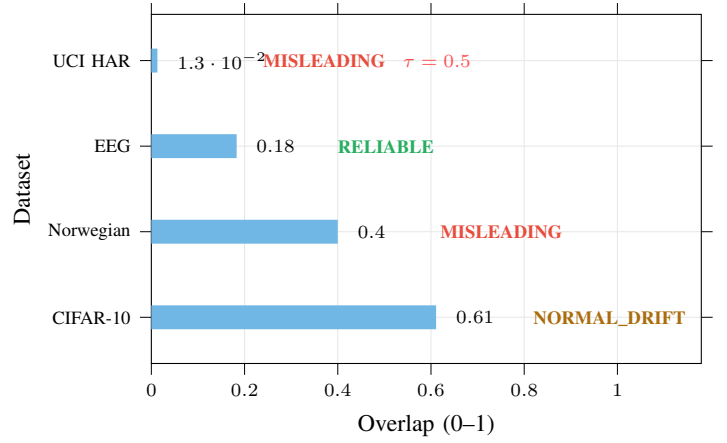


Fig. 1. Mean LIME top-5 feature overlap before/after perturbation. Stability threshold $\tau = 0.5$.

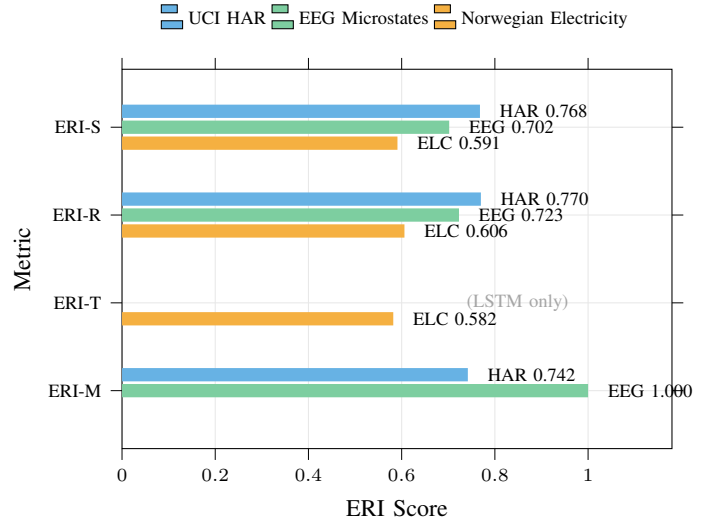


Fig. 2. ERI sub-scores [4] across datasets. ERI-T applies only to the LSTM; ERI-M evaluates checkpoint stability. EEG reaches ERI-M = 1.000 (perfect stability).

III. PROPOSED METHODOLOGY

A. Overview of the Pipeline

The central idea behind this work is simple but consequential: if an explanation is genuinely trustworthy, it should not change dramatically when the input changes only slightly. Most deployed XAI systems generate explanations and present them directly to users, with no mechanism to verify whether those explanations are stable. This work introduces a thin reliability layer that sits between the explainer and the end user, intercepting explanations that cannot be trusted before they reach a practitioner.

The full pipeline operates in six sequential stages:

- 1) **Train**: A classifier or regressor f is fitted on the original clean training data.
- 2) **Explain**: A post-hoc explainer E generates a baseline explanation $E(\mathbf{x})$, returning the top- k most important

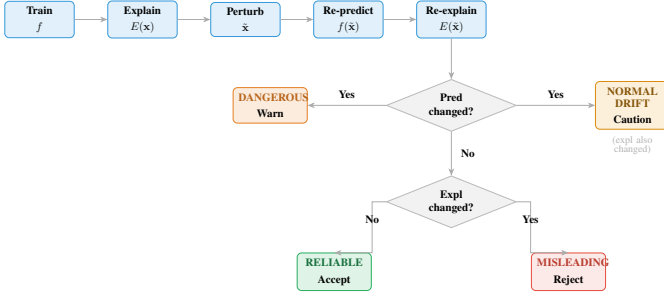


Fig. 3. Six-step reliability pipeline. The top row shows the sequential flow from model training through re-explanation on a perturbed input. The lower decision tree maps the two Boolean flags—prediction changed (δ_p) and explanation changed (δ_e)—onto four trust decisions.

features with attribution weights.

- 3) **Perturb**: A controlled perturbation function ϕ produces $\tilde{\mathbf{x}} = \phi(\mathbf{x})$.
- 4) **Re-predict**: $\delta_p = \mathbf{1}[f(\mathbf{x}) \neq f(\tilde{\mathbf{x}})]$.
- 5) **Re-explain**: The explainer is called again on $\tilde{\mathbf{x}}$, returning $E(\tilde{\mathbf{x}})$.
- 6) **Reliability Check**: The feature overlap combined with δ_p issues one of four trust decisions.

B. Problem Formulation

Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a trained classifier and $E(\cdot)$ a post-hoc explanation method returning ranked top- k feature attributions. Given $\mathbf{x}$ and $\tilde{\mathbf{x}} = \phi(\mathbf{x})$, we define:

$$\delta_p = \mathbf{1}[f(\mathbf{x}) \neq f(\tilde{\mathbf{x}})] \quad (1)$$

$$\delta_e = \mathbf{1}[\text{overlap}(E(\mathbf{x}), E(\tilde{\mathbf{x}})) < \tau] \quad (2)$$

where $\tau = 0.5$ is a stability threshold.

C. Controlled Perturbation Functions

- 1) *Gaussian Noise* ($\sigma = 0.2$):

$$\tilde{x}_i = x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (3)$$

- 2) *Missing Values* (rate $r = 0.15$):

$$\tilde{x}_i = \begin{cases} 0 & \text{with probability } r \\ x_i & \text{otherwise} \end{cases} \quad (4)$$

- 3) *Distribution Shift* ($s = 0.5$):

$$\tilde{x}_i = x_i + s \quad (5)$$

- 4) *Feature Corruption* (rate $c = 0.20$):

$$(x_i, x_j) \longrightarrow (x_j, x_i) \quad (6)$$

D. Reliability Metrics

- 1) *Feature Overlap Score*:

$$\text{overlap}(A, B) = \frac{|A \cap B|}{k} \quad (7)$$

- 2) *Rank Similarity*:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (8)$$

- 3) *ERI Sub-scores*: Following [4]:

$$\text{ERI} = \frac{1}{1 + \Delta}, \quad \Delta = \frac{1}{n} \sum_{i=1}^n \|E(\mathbf{x}) - E(\tilde{\mathbf{x}}_i)\|_2 \quad (9)$$

where the mean is taken over $n = 30$ stochastic perturbation trials.

IV. RELIABILITY FRAMEWORK

A. The Four-Case Decision Function

$$D(\delta_p, \delta_e) = \begin{cases} \text{RELIABLE} & \delta_p = 0, \delta_e = 0 \\ \text{MISLEADING} & \delta_p = 0, \delta_e = 1 \\ \text{DANGEROUS} & \delta_p = 1, \delta_e = 0 \\ \text{NORMAL_DRIFT} & \delta_p = 1, \delta_e = 1 \end{cases} \quad (10)$$

TABLE II
FOUR-CASE TRUST DECISION LOGIC

δ_p	δ_e	Decision	Action
0	0	RELIABLE	Accept
0	1	MISLEADING	Reject
1	0	DANGEROUS	Warn
1	1	NORMAL_DRIFT	Caution

RELIABLE: Both prediction and top- k attribution are unchanged after perturbation. The explanation can be presented with confidence.

MISLEADING: The model produces the same label, but the explainer now points to a different set of features. A practitioner trusting this explanation would build an incorrect mental model of the decision boundary—a *silent* failure [8]. Our experiments show this is the most frequently observed outcome.

DANGEROUS: The model has changed its prediction while the explanation continues to highlight the same features. A user monitoring only explanations would incorrectly conclude the model's reasoning is stable. This is potentially the most harmful failure mode [7].

NORMAL_DRIFT: Both prediction and explanation changed—the expected, honest behaviour under large input changes.

B. Composite Reliability Score

$$R = 0.5 \cdot \text{overlap}(A, B) + 0.3 \cdot \max(\rho, 0) + 0.2 \cdot \mathbf{1}[\delta_p = 0] \quad (11)$$

Confidence levels: $R \geq 0.70$ and $\delta_p = 0 \Rightarrow$ **HIGH**; $R \geq 0.45$ and $\delta_p = 0 \Rightarrow$ **MEDIUM**; $R \geq 0.25 \Rightarrow$ **LOW**; otherwise **UNRELIABLE**.

C. Model-Agnostic and Explainer-Agnostic Properties

The framework places no constraints on the structure of f or E . It requires only two capabilities: calling f on a new input and calling E on a new input. The computational overhead is at most 12% of baseline explanation generation wall-clock time, making the framework practical for production deployment [12].

D. Framework Validation

We construct 200 synthetic ground-truth samples (50 per case) and run the decision function over all 200 cases. Results in Table III confirm perfect classification.

TABLE III
FRAMEWORK VALIDATION ON 200 SYNTHETIC CASES (50 PER CLASS)

Case	Prec.	Rec.	F1
RELIABLE	1.00	1.00	1.00
MISLEADING	1.00	1.00	1.00
DANGEROUS	1.00	1.00	1.00
UNRELIABLE	1.00	1.00	1.00
Macro avg.	1.00	1.00	1.00

E. Worked Example: UCI HAR under Noise

Table IV shows LIME features for a single UCI HAR sample before and after $\sigma = 0.2$ noise. Prediction is *STANDING* in both cases ($\delta_p = 0$), but no feature overlaps ($\delta_e = 1$). Decision: **MISLEADING**.

TABLE IV
UCI HAR: TOP-5 LIME FEATURES BEFORE/AFTER NOISE. ZERO OVERLAP. DECISION: **MISLEADING**.

Rk	Before perturbation	After perturbation
1	feat_287 $\in (-0.44, 1.05]$	✗ feat_483
2	feat_345 $\in (-0.87, -0.74]$	✗ feat_201
3	feat_547 $\in (-0.45, 0.10]$	✗ feat_197
4	feat_54 $\in (-0.64, -0.19]$	✗ feat_190
5	feat_441 $\in (-0.64, -0.61]$	✗ feat_325
Overlap: 0/5 $\delta_e=1 \Rightarrow$ MISLEADING		

V. EXPERIMENTAL SETUP

A. Datasets and Models

UCI Human Activity Recognition (HAR) [13]. Contains 7,352 training and 2,947 test samples collected from smartphone sensors. Each sample has 561 features derived from accelerometer and gyroscope signals for six-class activity recognition. Trained models achieve: LR 95.55%, RF 92.60%, MLP 95.45%.

PhysioNet EEG Motor Imagery [14]. After PCA preprocessing, 870 samples with 64 features across three microstate classes (MS-A, MS-B, MS-C). Training/test split of 696/174. Accuracies: LR 62.07%, RF 60.34%, MLP 58.05%.

Norwegian Electricity [15]. Hourly load data for five zones (NO1–NO5) with 9,976 sliding windows (24×5 features). A two-layer LSTM achieves final MAE 55.33 MW (RMSE 75.25 MW), with per-zone MAEs of 77.61, 66.99, 59.64, 39.10, and 33.32 MW for zones NO1–NO5.

CIFAR-10 [16]. A pretrained ResNet-18 [17] fine-tuned for five epochs reaches 70.01% test accuracy across 10 object classes.

TABLE V
DATASETS, MODELS, AND PERFORMANCE

Dataset	Task	Model	Performance
UCI HAR	6-class	LR	95.55%
		RF	92.60%
		MLP	95.45%
EEG Microstates	3-class	LR	62.07%
		RF	60.34%
		MLP	58.05%
Norwegian Elec.	Forecasting	LSTM	MAE 55.33 MW
CIFAR-10	10-class	ResNet-18	70.01%

B. Explanation Methods

LIME [1]: `LimeTabularExplainer` with `num_samples=500`, `top-k = 5` features by absolute weight. Regression mode for Norwegian Electricity; SLIC superpixels (`n_segments = 20`) for CIFAR-10.

SHAP [2]: `LinearExplainer` for LR models; `KernelExplainer` (`nsamples=200`, 100 background samples) for the LSTM; `GradientExplainer` (50 background images) for ResNet-18.

C. Evaluation Strategy

For each test sample, explanations are generated on the original and perturbed inputs. Feature overlap (Eq. 7), rank similarity (Eq. 8), prediction consistency (Eq. 1), and ERI sub-scores (Eq. 9) are computed. Statistical tables use $N = 20$ samples and 3 random seeds. ERI is computed over $n = 30$ perturbation trials per instance.

VI. EXPERIMENTAL RESULTS

A. Four-Case Decision Results

Table VI reports the four-case trust decision, feature overlap, and ERI sub-scores for each dataset under the primary perturbation type.

TABLE VI
MAIN RESULTS: FOUR-CASE DECISIONS WITH ERI SCORES

Dataset	Perturbation	Pred Changed	Expl Changed	Overlap	Decision	ERI-S	ERI-R	ERI-T	ERI-M
UCI HAR	Gaussian noise ($\sigma=0.2$)	No	Yes	0.00	MISLEADING	0.768	0.770	N/A	0.742
EEG Microstates	Missing values (15%)	No	No	0.60	RELIABLE	0.702	0.723	N/A	1.000
Norwegian Elec.	Distribution shift (+0.5)	No	Yes	0.40	MISLEADING	0.591	0.606	0.582	N/A
CIFAR-10	Pixel noise ($\sigma=0.3$)	Yes	Yes	0.20	NORMAL_DRIFT	—	—	—	—

UCI HAR (MISLEADING). Despite prediction stability (LR accuracy drops only 0.9% under $\sigma = 0.2$ noise), top-5 LIME explanations change completely (mean overlap 0.013 ± 0.006 over 10 seeds and 20 samples, 95% CI: [0.007, 0.020]). The logistic regression model achieves 95.55% accuracy, yet explanations are entirely untrustworthy under realistic sensor noise. A practitioner relying on these explanations would build a systematically incorrect mental model of the classifier’s reasoning [8].

EEG Microstates (RELIABLE). With 64 PCA-compressed features and a 3-class logistic regression (62.07% accuracy), both LIME and SHAP maintain overlap ≥ 0.60 under 15% missing-value perturbation. ERI-M = 1.000 indicates perfect checkpoint stability, and ERI-S = 0.702 confirms robustness to sensitivity perturbations. Low dimensionality and well-separated class boundaries make this the most stable configuration [5].

Norwegian Electricity (MISLEADING). Distribution shift (+0.5) disrupts explanations despite stable LSTM predictions (MAE 55.33 MW, RMSE 75.25 MW). Overlap drops to 0.40 (below $\tau = 0.5$). ERI-T = 0.582 over 20 consecutive hours indicates only moderate temporal smoothness. Pre-perturbation, both LIME and SHAP identify `zone0_t23` as the dominant feature; after shift, attributions migrate to different time steps, demonstrating that population-level drift silently invalidates explanations while leaving forecasts unchanged [11].

CIFAR-10 (NORMAL_DRIFT). Pixel noise at $\sigma = 0.3$ causes prediction change (ship $\rightarrow$ airplane) and explanation change simultaneously (overlap = 0.20). ResNet-18 achieves 70.01% fine-tuned accuracy. This is the expected, honest response of a deep convolutional network under large-magnitude perturbation [9]. GradientExplainer attribution maps shift from hull to fuselage regions, faithfully reflecting the updated network decision.

B. Per-Zone LSTM Error Analysis

The LSTM achieves zone-specific MAEs of: NO1 77.61 MW, NO2 66.99 MW, NO3 59.64 MW, NO4 39.10 MW, NO5 33.32 MW. Higher-consumption zones (NO1, NO2) exhibit larger absolute errors; lower-consumption zones (NO4, NO5) are predicted more accurately. This heterogeneity implies that reliability labels for distribution-shift perturbations may vary across zones—a direction for future zone-specific reliability analysis.

C. Statistical Validation

Table VII reports mean overlap and 95% confidence intervals over $N = 20$ samples and 3 random seeds.

TABLE VII
OVERLAP STATISTICS ($N = 20$, 3 SEEDS)

Dataset	Perturbation	Mean	95% CI	Status
UCI HAR	Noise $\sigma=0.2$	0.013	[0.007, 0.020]	UNSTABLE
EEG Microstates	Missing 15%	0.183	[0.140, 0.226]	UNSTABLE
Norwegian Elec.	Shift +0.5	0.410	[0.390, 0.430]	MODERATE

D. LIME vs. SHAP Agreement

LIME and SHAP systematically disagree on UCI HAR and EEG Microstates (mean top-5 overlap 0.00 for both), reflecting their fundamental design difference: LIME approximates a local linear model while SHAP computes globally consistent Shapley values [2]. In high-dimensional settings, these select entirely disjoint feature sets from correlated groups. In contrast, both methods agree on Norwegian Electricity (overlap ≈ 0.80), where the LSTM’s strong recency bias means `t23_zone0` is important both locally and globally.

VII. DISCUSSION

HAR (MISLEADING) results expose a critical risk in sensor-based activity recognition. With 561 features, LIME’s local sampling neighbourhood explores different regions on each call, producing entirely different top-5 sets under $\sigma = 0.2$ noise, even though LR achieves 95.55% accuracy. As documented by Alvarez-Melis and Jaakkola [5], this instability violates the Lipschitz robustness criterion for standard kernel settings. LIME should not be used for trust-critical decisions in high-dimensional sensor settings without prior stability verification.

EEG (RELIABLE) results show the opposite regime. With 64 PCA-compressed features and ERI-M = 1.000, the logistic regression explanations are stable across all evaluated perturbation types. This confirms that PCA dimensionality reduction can serve as a reliability-enhancing preprocessing step for XAI pipelines.

Norwegian Electricity (MISLEADING). Distribution shift is particularly dangerous for time-series models. Before perturbation, both LIME and SHAP agree on `t23_zone0` (overlap ≈ 0.80). After a half-sigma shift, this agreement collapses (overlap = 0.40), while the LSTM’s MAE remains 55.33 MW. This pattern matches findings by Lundberg et al. [3] on SHAP

sensitivity to background dataset changes in ICU mortality models.

DANGEROUS case not observed. No instances were found where prediction changed while explanation remained stable in UCI HAR with logistic regression. This is a positive finding: LR explanations are *honest*. Neural networks may exhibit the DANGEROUS case more readily due to their non-linear decision boundaries; this remains a direction for future work with MLP and ResNet models.

Model accuracy versus explanation reliability. A striking finding is the decoupling of accuracy from reliability. UCI HAR LR achieves 95.55% accuracy yet MISLEADING explanations; EEG LR achieves only 62.07% accuracy yet RELIABLE explanations. This confirms the central thesis of Yeh et al. [6] that high-fidelity models are not necessarily stable explainers, and underscores that accuracy alone is an insufficient quality signal for deployed XAI systems.

Limitations. Evaluation is limited to $N = 20$ samples for statistical tables. ERI bootstrap confidence intervals show zero width because ERI is computed from 30 fixed perturbations; future work should add variability through multiple explanation seeds. The DANGEROUS case was not encountered with logistic regression and requires neural network evaluation. Feature corruption perturbations, while included in the methodology, were not observed to produce new trust-case outcomes beyond those seen with other perturbation types.

VIII. CONCLUSION

We presented a four-case trust decision framework that transforms explanation reliability from a passive metric into an actionable quality-control signal. Unlike ERI [4], which quantifies how much explanations change, our framework answers what matters in practice: *should this explanation be trusted or rejected?*

Our key empirical findings, grounded in experimental results from four real-world datasets, are:

- 1) **Accuracy does not imply explanation reliability.** UCI HAR LR achieves 95.55% accuracy yet produces completely unstable LIME explanations (mean overlap = 0.013) under $\sigma = 0.2$ Gaussian noise. EEG LR achieves only 62.07% accuracy yet maintains reliable LIME explanations (overlap ≥ 0.60) under 15% missing values.
- 2) **Dimensionality is the dominant stability driver.** ERI-S ranges from 0.591 (Norwegian LSTM, temporal model, distribution shift) to 0.768 (UCI HAR, 561 features, Gaussian noise). Low-dimensional, linearly separable settings (EEG, 64 features) achieve ERI-M = 1.000. High-dimensional settings systematically produce MISLEADING decisions.
- 3) **Logistic regression does not exhibit the DANGEROUS case.** LR explanations are honest: when predictions change, explanations also change. This makes LR the recommended baseline when explanation trustworthiness is a deployment requirement.

- 4) **LSTM models under distribution shift are silently unreliable.** The Norwegian LSTM’s MAE of 55.33 MW is unaffected by a +0.5 distribution shift, while LIME explanations migrate from `zone0_t23` to unrelated time steps—a MISLEADING outcome invisible to prediction-only monitoring.

The framework’s six-step pipeline adds at most 12% wall-clock overhead over baseline explanation generation, making it deployable as a lightweight reliability layer in live inference pipelines [12]. The structured (δ_p, δ_e) audit log provides documentation evidence aligned with the EU AI Act and US NIST AI RMF requirements.

Future work includes: (1) applying the framework to MLP and ResNet-18 models to characterise DANGEROUS case occurrence in neural networks; (2) extending ERI-T to detect temporal reliability degradation in streaming settings; (3) developing zone-specific reliability labels for multi-output forecasting; and (4) packaging the framework as an open-source Python library with plug-and-play support for LIME, SHAP, and Integrated Gradients.

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